SaTTric+ Monitoring Parameters and Utilities

Overview

SaTTric+ can act as an Energy Edge Box by combining external Modbus energy-meter data with analog, digital, and pulse-based field signals to create a broad monitoring layer for electrical, thermal, mechanical, utility, and infrastructure assets. The attached materials support this architecture through asset mapping across electrical, HVAC, water, gas, pressure, access, and utility systems, along with time-stamped energy, alarms, logging, and carbon-related data structures. The list below provides an expanded set of **more than 100 monitoring parameters** that can be viewed, logged, correlated, or used for edge decisions through SaTTric+. Parameters are grouped by domain, and each parameter includes a short statement of its practical utility.

Electrical and energy parameters

1. **Phase voltage L1-N** — Used to detect under-voltage, over-voltage, and phase health for phase L1.
2. **Phase voltage L2-N** — Used to verify voltage condition and stability on phase L2.
3. **Phase voltage L3-N** — Used to verify voltage condition and stability on phase L3.
4. **Line voltage L1-L2** — Used for line-to-line voltage assessment and three-phase system validation.
5. **Line voltage L2-L3** — Used to evaluate system balance and supply quality.
6. **Line voltage L3-L1** — Used to confirm three-phase symmetry and network condition.
7. **Current L1** — Used for feeder loading, overload detection, and current trend tracking on phase L1.
8. **Current L2** — Used for feeder loading, overload detection, and current trend tracking on phase L2.
9. **Current L3** — Used for feeder loading, overload detection, and current trend tracking on phase L3.

10. **Neutral current** — Used to detect imbalance, harmonic-rich loading, and neutral stress.
11. **Average current** — Used as a quick overall load indicator for incomers or large feeders.
12. **Real power L1** — Used to see active power contribution of phase L1.
13. **Real power L2** — Used to see active power contribution of phase L2.
14. **Real power L3** — Used to see active power contribution of phase L3.
15. **Total real power** — Used as the main demand parameter for energy management and load shedding.
16. **Apparent power L1** — Used to judge utilization and loading on phase L1.
17. **Apparent power L2** — Used to judge utilization and loading on phase L2.
18. **Apparent power L3** — Used to judge utilization and loading on phase L3.
19. **Total apparent power** — Used to assess supply-side capacity loading such as transformer utilization.
20. **Reactive power L1** — Used for identifying phase-wise reactive demand and compensation need.
21. **Reactive power L2** — Used for identifying phase-wise reactive demand and compensation need.
22. **Reactive power L3** — Used for identifying phase-wise reactive demand and compensation need.
23. **Total reactive power** — Used to assess overall correction needs for power factor improvement.
24. **Power factor L1** — Used to evaluate utilization quality and phase-wise inefficiency.
25. **Power factor L2** — Used to evaluate utilization quality and phase-wise inefficiency.
26. **Power factor L3** — Used to evaluate utilization quality and phase-wise inefficiency.
27. **Total power factor** — Used to identify overall PF penalties and avoidable system losses.

28. **Line frequency** — Used for supply stability monitoring and disturbance correlation.
29. **Imported active energy (kWh)** — Used as the principal utility consumption quantity for energy and carbon accounting.
30. **Exported active energy (kWh)** — Used where backfeed, captive generation, or recovery is relevant.
31. **Per-feeder or per-meter kWh** — Used for sub-load and process-block energy allocation.
32. **Demand kW** — Used for demand charge management and event-based limit control.
33. **Voltage THD** — Used to identify voltage distortion and power quality issues affecting sensitive assets.
34. **Current THD** — Used to detect harmonic-heavy loads such as VFDs, rectifiers, or nonlinear devices.
35. **Voltage sag flag or event count** — Used for disturbance analysis and root-cause tracing.
36. **Voltage swell flag or event count** — Used for detecting over-voltage stress events.
37. **Voltage unbalance** — Used to assess motor stress and phase asymmetry.
38. **Phase reversal or sequence status** — Used during commissioning and fault diagnostics.
39. **Alarm bitfield for electrical events** — Used as a compact representation of electrical abnormalities.
40. **Energy interval value (1-min or 15-min)** — Used for baseline, profile, and trend analysis.
41. **Interval CO2e** — Used to turn time-bucketed kWh into time-bucketed carbon records.
42. **Cumulative CO2e** — Used for asset-wise and meter-wise carbon accumulation over time.

43. **Emissions factor value** — Used to show which carbon conversion factor is currently active.
44. **Emissions factor profile ID or version** — Used for audit and reprehensibility of carbon calculations.
45. **Emissions factor validity window** — Used to ensure historical carbon values remain traceable to the right factor period.
46. **Device time / RTC timestamp** — Used as the master time base for all logged parameters and events.

Thermal, HVAC, and refrigeration parameters

47. **Ambient temperature** — Used to correlate site conditions with HVAC, cooling, and process energy.
48. **Ambient humidity** — Used to assess latent load and indoor environmental conditions.
49. **Room temperature** — Used for comfort or conditioned-space performance verification.
50. **Cold room temperature** — Used for refrigeration compliance and energy-vs-temperature trade-off evaluation.
51. **Supply air temperature** — Used to evaluate actual delivered HVAC condition.
52. **Return air temperature** — Used to evaluate air-side thermal performance and temperature difference.
53. **Duct static pressure** — Used for fan control optimization and over-pressurization detection.
54. **Filter differential pressure** — Used to detect choked filters and rising fan energy demand.
55. **Coil differential pressure** — Used to identify hydraulic or air-side restriction issues.
56. **Chilled-water supply temperature** — Used to evaluate cooling-system delivery effectiveness.

57. **Chilled-water return temperature** — Used to calculate thermal extraction and ΔT utilization.
58. **Hot-water supply temperature** — Used to assess heating-side delivery.
59. **Hot-water return temperature** — Used to detect bypassing, low load, or re circulation losses.
60. **Suction pressure (refrigeration)** — Used for COP-related analysis and leakage or drift indications.
61. **Discharge pressure (refrigeration)** — Used to assess compressor loading and condenser behavior.
62. **Suction temperature (refrigeration)** — Used for system efficiency and operating-state analysis.
63. **Discharge temperature (refrigeration)** — Used to detect abnormal compression or high condensing stress.
64. **Defrost status** — Used to relate refrigeration energy to defrost behavior and optimization opportunity.
65. **Door contact status for conditioned zones** — Used to correlate energy waste with cold-room or controlled-zone access behavior.

Pressure, flow, and utility process parameters

66. **Water flow** — Used to compute specific energy per cubic meter in pumping and treatment systems.
67. **Chilled-water flow** — Used to relate pumping energy to actual thermal transport.
68. **Hot-water flow** — Used to measure heating utility throughput and intensity.
69. **Gas flow (analog)** — Used for combustion-system and utility accounting.
70. **Steam flow (analog)** — Used for steam allocation and utility performance analysis.
71. **Fuel flow** — Used for generators, burners, or transfer pumps where liquid fuel is measured.
72. **Booster pressure** — Used to avoid over-pressure and excessive pumping energy.

73. **Pipe differential pressure** — Used to tune pump speed, valve strategy, and circulation efficiency.
74. **Tank level** — Used for pump control logic, low-level protection, and storage tracking.
75. **Chemical dosing level or analog concentration signal** — Used to connect treatment intensity with utility conditions.
76. **Pressurisation unit status/pressure** — Used for utility-system stability and energy-efficient circulation.

Pulse-based utility and legacy integration parameters

77. **Water meter pulse count** — Used to totalize water consumption from legacy meters.
78. **Gas meter pulse count** — Used to totalize gas consumption and support Scope 1 related monitoring.
79. **Steam pulse count** — Used to totalize steam usage where pulse-output totalizers exist.
80. **Fuel pulse count** — Used to measure generator or boiler fuel usage from pulse-output sources.
81. **Production pulse count** — Used to relate utilities to production quantity or machine cycles.
82. **Runtime pulse count** — Used to measure operating hours from legacy counters.
83. **Chemical dosing pulse count** — Used to totalize dosing activity and relate it to water treatment behavior.

Mechanical, equipment-state, and safety parameters

84. **Pump run status** — Used to confirm operation and correlate runtime with kWh and process need.
85. **Pump trip status** — Used for event logging and reliability diagnosis.
86. **Fan run status** — Used for occupancy and schedule validation in HVAC systems.

87. **Fan trip status** — Used to identify loss of ventilation or abnormal equipment stoppage.
88. **Compressor run status** — Used to measure loading and operating behavior in refrigeration or air systems.
89. **Compressor trip or fault status** — Used to correlate faults with energy spikes, pressure issues, or maintenance events.
90. **Burner permissive or run status** — Used to understand thermal plant sequence and demand response.
91. **Valve open/closed status** — Used to verify hydraulic or thermal path behavior.
92. **Damper position proof or status** — Used to validate airflow path and ventilation logic.
93. **Filter dirty alarm** — Used to initiate maintenance before energy rises excessively.
94. **Occupancy status** — Used for lighting, HVAC, and auxiliary control logic.[file:3]
95. **Door access status** — Used to relate human activity to utility use and after-hours consumption.
96. **Panel door open status** — Used for electrical room or panel event traceability.
97. **Fire alarm contact** — Used to log emergency conditions and drive safe response logic.
98. **Gas detection alarm** — Used to trigger safety and ventilation response paths.
99. **UPS on-battery status** — Used to separate backup events from normal utility operation.
100. **UPS bypass/online status** — Used to understand actual UPS operating mode.
101. **Generator available/run status** — Used to track standby generation state and event windows.
102. **Lift status** — Used to isolate vertical-transport behavior from other building loads.

103. **Barrier or gate status** — Used to track powered access devices as auxiliary utility consumers.

104. **Lighting circuit ON/OFF status** — Used to validate scheduling, occupancy logic, and after-hours energy control.

Condition monitoring and derived utility indicators

105. **Vibration RMS** — Used for rotating-equipment health and energy-vs-condition correlation.

106. **Vibration peak value** — Used to detect impacts, looseness, or abnormal rotating behavior.

107. **Acoustic level or sound feature** — Used for anomaly detection such as cavitation, chatter, or unusual machine behavior.

108. **Specific energy per unit water** — Used as a derived KPI for pumping and treatment systems.

109. **Specific energy per unit production** — Used to link utilities directly to production output where pulse counts are available.

110. **Specific energy per runtime hour** — Used to compare machine efficiency over operating periods.

111. **After-hours consumption indicator** — Used to detect avoidable nonproductive energy use.

112. **Abnormal baseload indicator** — Used to flag persistent hidden energy waste during nominal idle conditions.

113. **Asset-level energy allocation tag** — Used to connect measured energy to utility blocks, process areas, or support systems.

114. **Asset-level carbon allocation tag** — Used to connect CO2e to feeder, utility, or production-related boundaries.

Practical note

Not every site will use all parameters at the same time. The practical strength of SaTTric+ is that it can combine meter parameters, sensor parameters, pulse parameters, equipment states, and derived KPIs in one timestamped edge architecture so that the same platform can serve electrical monitoring, utility optimization, digital-twin modeling, and CBAM-related attribution requirements.